Chapter 2
RANDOM VARIABLES AND PROBABILITY DISTRIBUTIONS

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CHAPTER OUTLINE

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LEARNING OBJECTIVES

After careful study of this chapter you should be able to do the following:

(1) Discrete Random Variables and Probability Distributions

1. Determine probabilities from probability mass functions and the reverse.
2. Determine probabilities from cumulative distribution functions and cumulative distribution functions from probability mass functions, and the reverse.
3. Calculate means and variances for discrete random variables.
4. Understand the assumptions for some common discrete probability distributions.
5. Select an appropriate discrete probability distribution to calculate probabilities in specific applications.
6. Calculate probabilities, determine means and variances for some common discrete probability distributions.

LEARNING OBJECTIVES

(2) Continuous Random Variables and Probability Distributions

1. Determine probabilities from probability density functions.
2. Determine probabilities from cumulative distribution functions and cumulative distribution functions from probability density functions, and the reverse.
3. Calculate means and variances for continuous random variables.
4. Understand the assumptions for some common continuous probability distributions.
5. Select an appropriate continuous probability distribution to calculate probabilities in specific applications.
6. Calculate probabilities, determine means and variances for some common continuous probability distributions.
7. Standardize normal random variables.
8. Use the table for the cumulative distribution function of a standard normal distribution to calculate probabilities.
9. Approximate probabilities for some binomial and Poisson distributions.

CONTENT

Random Experiments

Definition 1 (Random variable)

A random variable is a function that assigns a real number to each outcome in the sample space of a random experiment.

✍ A random variable is denoted by an uppercase letter such as X. After an experiment is conducted, the measured value of the random variable is denoted by a lowercase letter such as $x = 70$ milliamperes.

Random Experiments

Example 1

Here are some random variables:

1. X, the number of students asleep in the next probability lecture.
2. Y, the number of phone calls you answer in the next hour.
3. Z, the number of minutes you wait until you next answer the phone.

✍ Random variables X and Y are discrete random variables. The possible values of these random variables form a countable set. The underlying experiments have sample spaces that are discrete. The random variable Z can be any nonnegative real number. It is a continuous random variable. Its experiment has a continuous sample space.

Discrete Random Variable

Definition 2 (Discrete random variable)

A discrete random variable is a random variable with a finite (or countably infinite) range.

✍ Hence,

1. X is a finite random variable if the range is a finite set
$$S_X = \{x_1, x_2, \ldots, x_n\}.$$
2. X is a discrete random variable if the range of X is a countable set
$$S_X = \{x_1, x_2, \ldots\}.$$

Discrete Random Variable

Example 2

A voice communication system for a business contains 48 external lines. At a particular time, the system is observed, and some of the lines are being used. Let the random variable X denote the number of lines in use. Then, X can assume any of the integer values 0 through 48,

$$S_X = \{0, 1, \ldots, 48\}.$$

When the system is observed if 10 lines are in use, $x = 10$.

Continuous Random Variable

Definition 3 (Continuous random variable)

A continuous random variable is a random variable with an interval (either finite or infinite) of real numbers for its range.

Example 3

Let Y be the random variable defined by the waiting time, in hours, between successive speeders spotted by a radar unit. The random variable Y takes on all values y for which $y \geq 0$. Y is a continuous random variable.

Functions of a Random Variable

Definition 4 (Mathematical function)

Each sample value y of a derived random variable Y is a mathematical function $g(x)$ of a sample value x of another random variable X. We adopt the notation $Y = g(X)$ to describe the relationship of the two random variables.

Example 4

In Example 2, the random variable X denotes the number of lines in use, the square of the number of lines in use is X^2 and

$$S_{X^2} = \{0^2, 1^2, 2^2, \dots, 48^2\}.$$

Functions of a Random Variable

Example 5

The random variable X is the number of pages in a facsimile transmission. The phone company offers you a new charging plan for faxes: \$0.10 for the first page, \$0.09 for the second page, etc., down to \$0.06 for the fifth page. For all faxes between 6 and 10 pages, the phone company will charge \$0.50 per fax. (It will not accept faxes longer than ten pages.) The function $Y = g(X)$ for the charge in cents for sending one fax is

$$Y = g(X) = \begin{cases} 10.5X - 0.5X^2, & 1 \le X \le 5, \\ 50, & 6 \le X \le 10. \end{cases}$$

CONTENT

Probability Mass Functions

Definition 5 (Probability mass function)

For a discrete random variable X with $S_X = \{x_1, x_2, \ldots, x_n\}$, a probability mass function (PMF) is a function $P_X(x)$ such that

1. $P_X(x_i) = P(X = x_i)$ for all $i = 1, 2, \ldots, n$;
2. $P_X(x_i) \geq 0$ for all $i = 1, 2, \ldots, n$;
3. $\sum_{i=1}^{n} P_X(x_i) = 1$.

Probability Mass Functions

Example 6

There is a chance that a bit transmitted through a digital transmission channel is received in error. The chance that a bit transmitted through a digital transmission channel is received in error is 0.1. Also, assume that the transmission trials are independent. Let X be the number of bits in error in the next four bits transmitted. Find the PMF, $P_X(x)$, of X.

Solution.

- $P_X(0) = 0.6561$, $P_X(1) = 0.2916$, $P_X(2) = 0.0486$, $P_X(3) = 0.0036$, $P_X(4) = 0.0001$. **Why?**
- Check that the probabilities sum to 1.

A graphical description of the probability distribution of X is shown in Fig. 1.

Probability Mass Functions

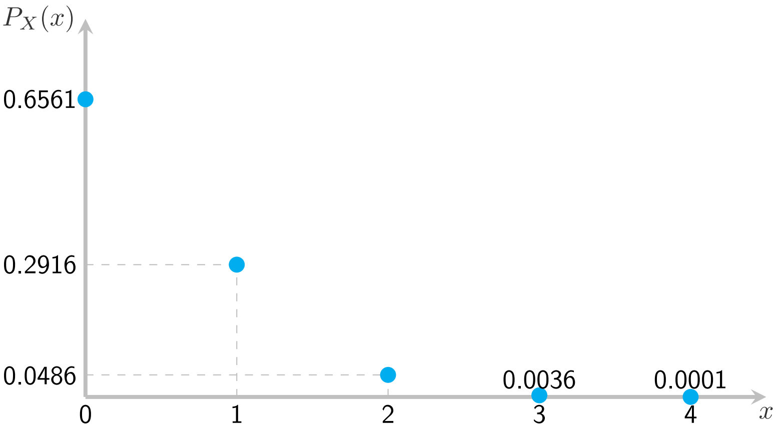


Figure 1: Probability distribution for bits in error.

Probability Distribution

Definition 6 (Probability Distribution)

The probability distribution for a discrete random variable X is a formula, table, or graph that gives the possible values of X, and the probability associated with each value of X.

X	x_1	x_2	$\dots$	x_n
$P_X(x_i)$	$P_X(x_1)$	$P_X(x_2)$	$\dots$	$P_X(x_n)$

(1)

✍ Requirements for discrete probability distribution:

1. The probability of each value of the discrete random variable is between 0 and 1, inclusive ($0 \leq P_X(x_i) \leq 1,\ i = 1, 2, \dots, n$).
2. The sum of all the probabilities is 1, that is $\sum_{i=1}^{n} P_X(x_i) = 1$.

Probability Distribution

Example 7

In Example 2, X is the number of bits in error in the next four bits transmitted. The probability distribution of X is

X	0	1	2	3	4
$P_X(x_i)$	0.6561	0.2916	0.0486	0.0036	0.0001

Probability Distribution

Example 8

In Example 5, suppose all your faxes contain $1, 2, 3$, or 4 pages with equal probability. Find the PMF of Y, the charge for a fax.

Solution.

$$P_Y(y) = \begin{cases} 1/4, & x = 10, 19, 27, 34, \\ 0, & \text{otherwise.} \end{cases}$$

Why?

Probability Distribution

Example 9

A shipment of 30 similar laptop computers to a retail outlet contains 5 that are defective. If a school makes a random purchase of 3 of these computers, find the probability distribution for the number of defectives.

Solution. Let X be a random variable whose values x are the possible numbers of defective computers purchased by the school. The probability distribution of X is

X	0	1	2	3
$P_X(x_i)$	$\frac{115}{203}$	$\frac{75}{203}$	$\frac{25}{406}$	$\frac{1}{406}$

Why?

Probability Distribution

Example 10

The probability that a bit transmitted through a digital transmission channel is received in error is 0.001. Assume the transmissions are independent events, and let the random variable X denote the number of bits transmitted until the first error. Find the probability distribution for X.

Solution. The probability distribution of X is

X	1	2	3	...	n	...
$P_X(x_i)$	0.001	0.999×0.001	$(0.999)^2 \times 0.001$	...	$(0.999)^{n-1} \times 0.001$	...

Why?

Probability Distribution

Theorem 1

For a discrete random variable X with PMF $P_X(x)$ and range S_X. If $B \subset S_X$, the probability that X is in the set B is

$$\boxed{P(B) = \sum_{x \in B} P_X(x)} \tag{2}$$

Example 11

In Example 6, we might be interested in the probability of fewer than four bits being in error. This question can be expressed as $P(X < 4)$. The event that is the union of the events $(X = 0)$, $(X = 1)$, $(X = 2)$, and $(X = 3)$. Clearly, these three events are mutually exclusive. Therefore,

$$P(X < 4) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) = 0.9999.$$

This approach can also be used to determine $P(X = 3) = P(X < 4) - P(X < 3) = 0.0036$.

Cumulative Distribution Function

Definition 7 (Cumulative distribution function)

The cumulative distribution function (CDF) $F_X(x)$ of a random variable X is

$$\boxed{F_X(x) = P(X < x) \quad \text{for } -\infty < x < \infty.} \tag{3}$$

- If X is a discrete random variable with probability distribution $P_X(x)$, then the CDF is $F_X(x) = \sum_{t<x} P_X(t)$.
- If X is a discrete random variable with probability distribution is (1), then the CDF is

$$\boxed{F_X(x) = \begin{cases} 0, & x \le x_1, \\ P_X(x_1), & x_1 < x \le x_2, \\ P_X(x_1) + P_X(x_2), & x_2 < x \le x_3, \\ \dots & \\ 1, & x > x_n. \end{cases}} \tag{4}$$

Cumulative Distribution Function

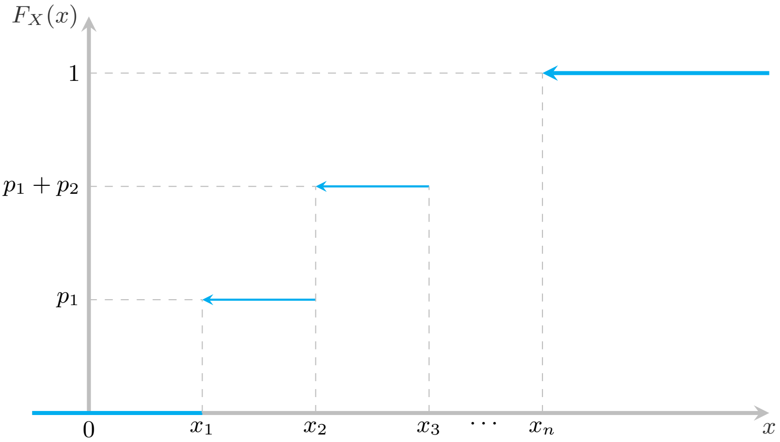


Figure 2: Figure of the CDF

Cumulative Distribution Function

Example 12

The CDF of random variable X in Example 9 is

$$F_X(x) = \begin{cases} 0, & x \leq 0, \\ \frac{115}{203}, & 0 < x \leq 1, \\ \frac{115}{203} + \frac{75}{203}, & 1 < x \leq 2, \\ \frac{115}{203} + \frac{75}{203} + \frac{25}{406}, & 2 < x \leq 3, \\ \frac{115}{203} + \frac{75}{203} + \frac{25}{406} + \frac{1}{406}, & x > 3. \end{cases} = \begin{cases} 0, & x \leq 0, \\ \frac{115}{203}, & 0 < x \leq 1, \\ \frac{190}{203}, & 1 < x \leq 2, \\ \frac{405}{406}, & 2 < x \leq 3, \\ 1, & x > 3. \end{cases}$$

Cumulative Distribution Function

Theorem 2

For any discrete random variable X with range $S_X = \{x_1, x_2, \dots\}$ satisfying $x_1 \le x_2 \le \dots$,

1. $0 \le F_X(x) \le 1$ for all x.
2. For all $x_1 < x_2$, $F_X(x_1) \le F_X(x_2)$ and $\lim_{x \to a^-} F_X(x) = F_X(a)$ for all $a \in \mathbb{R}$.
3. $F_X(-\infty) = 0$ and $F_X(+\infty) = 1$.
4. For $x_i \in S$ and ε, an arbitrarily small positive number, $F_X(x_i + \varepsilon) - F_X(x_i) = P_X(x_i)$.
5. $F_X(x) = F_X(x_i)$ for all x such that $x_i < x \le x_{i+1}$.
6. For all $b > a$,

$$\boxed{F_X(b) - F_X(a) = P(a \le X < b).} \tag{5}$$

CONTENT

Cumulative Distribution Function

Definition 8 (Continuous random variable)

X is a continuous random variable if the CDF $F_X(x) = P(X < x),\ x \in \mathbb{R}$ is a continuous function.

Remark 1

If X is a continuous random variable, then

$$\boxed{P(X = a) = 0.} \tag{6}$$

Follows directly from $P(a \le X < b) = F_X(b) - F_X(a)$ and $F_X(x)$ is a continuous function. Therefore, for all $b > a$,

$$\boxed{P(a \le X < b) = P(a < X \le b) = P(a \le X \le b) = P(a < X < b) = F_X(b) - F_X(a).} \tag{7}$$

This is not true in general for discrete random variables.

Cumulative Distribution Function

Example 13

Suppose the cumulative distribution function of the continuous random variable X is

$$F_X(x) = \begin{cases} 0, & x \leq 0, \\ A + 3x^2 - Bx^3, & 0 < x \leq 1, \\ 1, & x > 1. \end{cases}$$

(a) Find A and B. (b) Find $P(0.5 < X < 1)$.

Cumulative Distribution Function

Solution.

(a) $A = 0$, $B = 2$ and $F_X(x) = \begin{cases} 0, & x \leq 0, \\ 3x^2 - 2x^3, & 0 < x \leq 1, \\ 1, & x > 1. \end{cases}$

(b) $P(0.5 < X < 1) = 0.5$

Why?

Cumulative Distribution Function

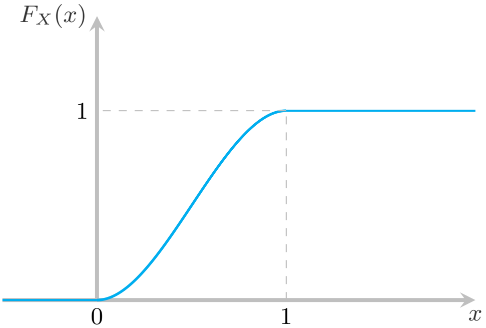


Figure 3: The CDF of Example 13

Cumulative Distribution Function

Example 14

Let the continuous random variable X denote the diameter of a hole drilled in a sheet metal component. The target diameter is 12.5 millimeters. Most random disturbances to the process result in larger diameters. Historical data show that the distribution of X can be modeled by a cumulative distribution function

$$F_X(x) = \begin{cases} 0, & x \leq 12.5, \\ 1 - e^{-20(x+k)}, & x > 12.5. \end{cases}$$

(a) Find k. (b) If a part with a diameter larger than 12.60 millimeters is scrapped, what proportion of parts is scrapped?

Cumulative Distribution Function

Solution.

(a) $k = -12.5$ and

$$F_X(x) = \begin{cases} 0, & x \le 12.5, \\ 1 - e^{-20(x-12.5)}, & x > 12.5. \end{cases}$$

(see Figure 4).

(b) $P(X > 12.6) \simeq 0.1353.$

Why?

Cumulative Distribution Function

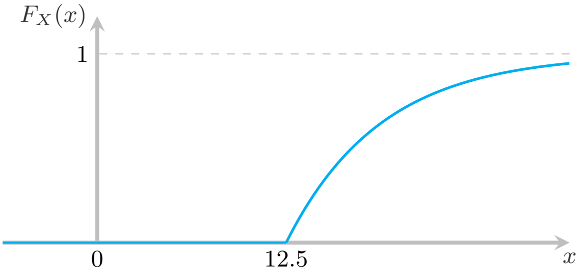


Figure 4: The CDF of Example 14

Cumulative Distribution Function

Example 15

The CDF of the continuous random variable X has form

$$F_X(x) = A + B \arctan x, \quad -\infty < x < +\infty.$$

(a) Find A and B. (b) Find the probability that when observing the random variable X three times independently, there are two times when X takes the value in the interval $(-1; 1)$.

Solution. (a) $A = {}^{1}/_{2}$, $B = {}^{1}/_{\pi}$ and

$$F_X(x) = \frac{1}{2} + \frac{1}{\pi} \arctan x, \quad -\infty < x < +\infty.$$

(b) $P_3(2) = 3/8$.
Why?

Probability Density Function

Definition 9 (Probability density function)

The probability density function (PDF) of a continuous random variable X is

$$\boxed{f_X(x) = \frac{dF_X(x)}{dx}.} \quad (8)$$

Theorem 3

For a continuous random variable X with PDF $f_X(x)$,

1. $f_X(x) \geq 0$ for all x,
2. $\displaystyle\int_{-\infty}^{+\infty} f_X(x)dx = 1$,
3. $\displaystyle F_X(x) = \int_{-\infty}^{x} f_X(u)du$.

Probability Density Function

Theorem 4

$$\boxed{P(a < X < b) = \int_a^b f_X(x)dx.} \tag{9}$$

Probability Density Function

✍ The probability that X will fall into a particular interval say, from a to b is equal to the area under the curve between the two points a and b. This is the shaded area in Figure 5.

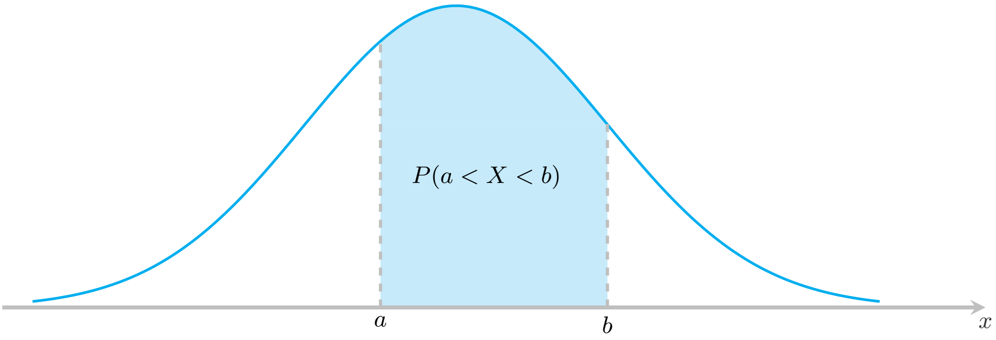


Figure 5: The probability distribution $f_X(x)$; $P(a < X < b)$ is equal to the shaded area under the curve

Probability Density Function

Example 16

Let the continuous random variable X denote the current measured in a thin copper wire in milliamperes. Assume that the range of X is $[0,\ 20\ \text{mA}]$, and assume that the probability density function of X is

$$f_X(x) = \begin{cases} 0, & x \notin [0; 20], \\ 0.05, & x \in [0; 20]. \end{cases}$$

What is the probability that a current measurement is less than 10 milliamperes?

Solution. $P(X < 10) = 0.5$

Why?

Probability Density Function

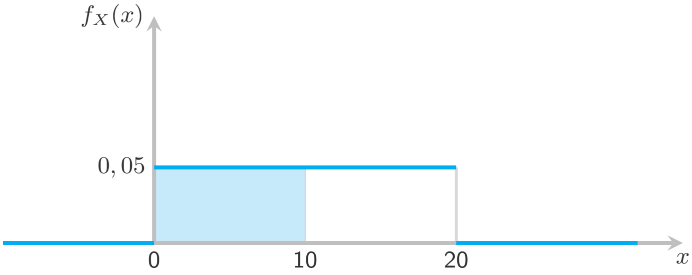


Figure 6: Probability density function for Example 16

Probability Density Function

Example 17

Let the continuous random variable X denote the diameter of a hole drilled in a sheet metal component. The target diameter is 12.5 millimeters. Most random disturbances to the process result in larger diameters. Historical data show that the distribution of X can be modeled by a probability density function

$$f_X(x) = \begin{cases} 0, & x \le 12.5, \\ 20e^{-20(x-12.5)}, & x > 12.5. \end{cases}$$

(a) If a part with a diameter larger than 12.60 millimeters is scrapped, what proportion of parts is scrapped? (b) What proportion of parts is between 12.5 and 12.6 millimeters? (c) Find $F_X(x)$.

Probability Density Function

Solution.

(a) $P(X > 12.6) \simeq 0.1353.$

(b) $P(12.5 < X < 12.6) = 0.865.$

(c) $F_X(x) = \begin{cases} 0, & x \leq 12.5, \\ 1 - e^{-20(x-12.5)}, & x > 12.5. \end{cases}$

Why?

Probability Density Function

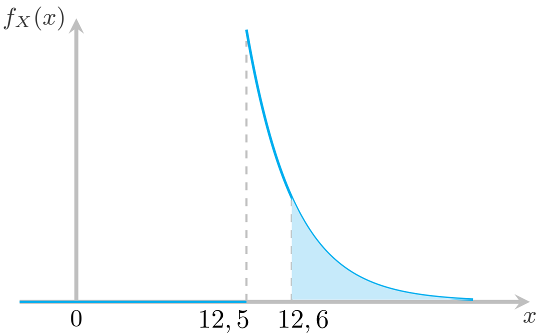


Figure 7: Probability density function for Example 17

Probability Density Function

Example 18

The continuous random variable X has PDF

$$f_X(x) = ae^{-|x|}, \quad -\infty < x < +\infty.$$

Define the random variable Y by $Y = X^2$. (a) What is a? (b) What is the CDF $F_X(x)$? (c) What is the CDF $F_Y(x)$? (d) Find $P(0 < X < \ln 3)$?

Probability Density Function

Solution.

(a) $a = \frac{1}{2}$.

(b) $$F_X(x) = \begin{cases} \dfrac{1}{2}e^{x}, & \text{if } \quad x \leq 0, \\ 1 - \dfrac{1}{2}e^{-x}, & \text{if } \quad x > 0. \end{cases}$$

(c) $$F_Y(x) = \begin{cases} 0, & \text{if } \quad x \leq 0, \\ 1 - e^{-\sqrt{x}}, & \text{if } \quad x > 0. \end{cases}$$

(d) $P(0 < X < \ln 3) = 1/3$.

Why?

CONTENT

Mode and Median

Definition 10 (Mode)

A mode of random variable X, $\text{mod}(X)$, is a number x_{mod} satisfying

$$\boxed{P_X(x_{\text{mod}}) \geq P_X(x) \quad \text{for all} \quad x} \quad \text{if } X \text{ is discrete} \tag{10}$$

or

$$\boxed{f_X(x_{\text{mod}}) \geq f_X(x) \quad \text{for all} \quad x} \quad \text{if } X \text{ is continuous.} \tag{11}$$

Mode and Median

Definition 11 (Median)

A median med(X), x_{med}, of random variable X is a number that satisfies

$$\boxed{P(X < x_{\text{med}}) = P(X \geq x_{\text{med}})} \tag{12}$$

Example 19

The probability density function of the continuous random variable X is

$$f_X(x) = \begin{cases} \frac{3}{4}x(2-x), & 0 \leq x \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

What is x_{mod}? What is x_{med}?

Mode and Median

Solution. $x_{\text{med}} = 1$ and $x_{\text{mod}} = 1$.

Why?

Expected Value of a Random Variable

Definition 12 (Expected value/Mean value)

Let X be a random variable with probability distribution $P_X(x)$ or $f_X(x)$. The mean value or expected value of X, denoted as μ_X or $E(X)$, is

$$\boxed{E(X) = \sum_{x \in S_X} x P_X(x)} \quad \text{if } X \text{ is discrete} \tag{13}$$

and

$$\boxed{E(X) = \int_{-\infty}^{+\infty} x f_X(x) dx} \quad \text{if } X \text{ is continuous.} \tag{14}$$

Expected Value of a Random Variable

Example 20

(a) In Example 6, there is a chance that a bit transmitted through a digital transmission channel is received in error. Let X equal the number of bits in error in the next four bits transmitted. Now $E(X)$ is

$$E(X) = (0)(0.6561) + (1)(0.2916) + (2)(0.0486) + (3)(0.0036) + (4)(0.0001) = 0.4.$$

(b) In Example 10, $E(X)$ is

$$E(X) = 1 \times 0.001 + 2 \times 0.999 \times 0.001 + 3 \times (0.999)^2 \times 0.001 + \cdots = 0.001 \times \sum_{n=1}^{\infty} n \times (0.999)^{n-1} = 1000.$$

(c) For the copper current measurement in Example 16, the mean of X is

$$E(X) = \int_0^{20} x f_X(x)dx = \int_0^{20} 0.05x dx = 10.$$

Expected Value of a Function of a Random Variable

Theorem 5

Let X be a random variable with probability distribution $P_X(x)$ or $f_X(x)$. The expected value of the random variable $Y = g(X)$ is

$$\boxed{E[g(X)] = \sum_{x \in S_X} g(x) P_X(x)} \quad \text{if } X \text{ is discrete} \tag{15}$$

and

$$\boxed{E[g(X)] = \int_{-\infty}^{+\infty} g(x) f_X(x) dx} \quad \text{if } X \text{ is continuous.} \tag{16}$$

Expected Value of a Function of a Random Variable

Example 21

In Example 5, suppose all your faxes contain $1, 2, 3$, or 4 pages with equal probability. Find the expected value of Y, the charge for a fax.

Solution.

- Using the result of Example 8, $E(Y) = 22.5$
- We can verify this result by applying Theorem 5, $E(Y) = 22.5$.

Why?

Expected Value of a Random Variable

Example 22

(a) In Example 6, let X equal the number of bits in error in the next four bits transmitted. Using (15),

$$E(X^2) = 0^2 \times 0.6561 + 1^2 \times 0.2916 + 2^2 \times 0.0486 + 3^2 \times 0.0036 + 4^2 \times 0.0001 = 0.52.$$

Practical Interpretation: The expected value of a function of a random variable is simply a weighted average of the function evaluated at the values of the random variable.

(b) In Example 16, the continuous random variable X denote the current measured in a thin copper wire in milliamperes. Using (16),

$$E(X^2) = \int_0^{20} x^2 f_X(x)dx = \int_0^{20} 0.05x^2 dx = 133.3333.$$

Properties of Expected Value

Theorem 6

For any random variable X,

1. $E[X - E(X)] = 0$.
2. $E(aX + b) = aE(X) + b$.

Corollary 1

1. Setting $a = 0$, we see that $E(b) = b$.
2. Setting $b = 0$, we see that $E(aX) = aE(X)$.

Properties of Expected Value

Example 23

Find the expected value of $Y = 6X + 2$, where X in Example 9.

Solution.

- Using (13) and Theorem 6, $E(6X + 2) = 5$.
- We can verify this result by applying Theorem 5, $E(6X + 2) = 5$.

Why?

Properties of Expected Value

Theorem 7

The expected value of the sum or difference of two or more functions of a random variable X is the sum or difference of the expected values of the functions. That is,

$$E[g(X) \pm h(X)] = E[g(X)] \pm E[h(X)]. \quad (17)$$

Example 24

Let X be a random variable with probability distribution as follows:

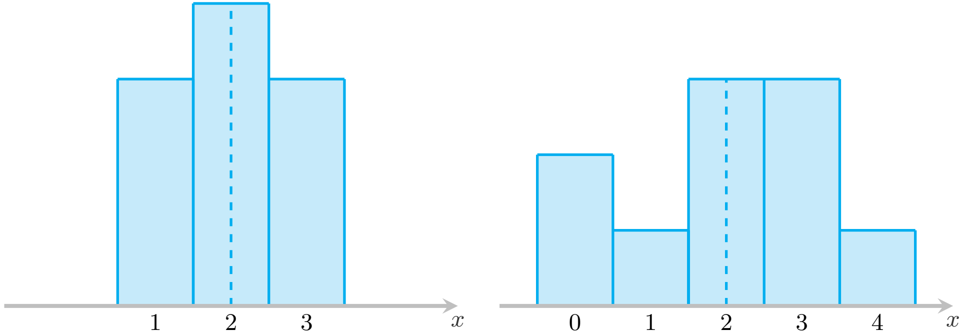
X	0	1	2	3
$P(X = x_i)$	1/3	1/2	0	1/6

Find the expected value of $Y = (X-1)^2$.

Solution. $E(X-1)^2 = 1$.
Why?

Variance and Standard Deviation

✍ The mean, or expected value, of a random variable X is of special importance in statistics because it describes where the probability distribution is centered. By itself, however, the mean does not give an adequate description of the shape of the distribution. We also need to characterize the variability in the distribution. In Figure 8, we have the histograms of two discrete probability distributions that have the same mean, $\mu = 2$, but differ considerably in variability, or the dispersion of their observations about the mean.



(a) (b)

Variance of a Rondom Variable

✍ The most important measure of variability of a random variable X is obtained by applying Theorem 5 with $g(X) = [X - E(X)]^2$. The quantity is referred to as the variance of the random variable X or the variance of the probability distribution of X and is denoted by Var(X) or the symbol σ_X^2, or simply by σ^2.

Definition 13 (Variance)

The variance of random variable X is

$$\boxed{\mathrm{Var}(X) = E\left[(X - E(X))^2\right] = \sum_x (x-\mu)^2 P_X(x) \quad \text{if } X \text{ is discrete}} \tag{18}$$

and

$$\boxed{\mathrm{Var}(X) = E\left[(X - E(X))^2\right] = \int\limits_{-\infty}^{+\infty} (x-\mu)^2 f_X(x)dx \quad \text{if } X \text{ is continuous.}} \tag{19}$$

Variance of a Rondom Variable

Definition 14 (Standard deviation)

The positive square root of the variance, σ_X or σ, is called the standard deviation of X.

$$\sigma_X = \sqrt{\mathrm{Var}(X)} \tag{20}$$

Variance of a Rondom Variable

Example 25

Let the random variable X represent the number of automobiles that are used for official business purposes on any given workday. The probability distribution for company A and that for company B (Figure 8) are

X_A	1	2	3
$P(X_A = x_i)$	0.3	0.4	0.3

X_B	0	1	2	3	4
$P(X_B = y_j)$	0.2	0.1	0.3	0.3	0.1

Show that the variance of the probability distribution for company B is greater than that for company A.

Variance of a Rondom Variable

Solution. For company A and B, we find that

$$E(X_A) = 1 \times 0.3 + 2 \times 0.4 + 3 \times 0.3 = 2.0,$$
$$E(X_B) = 0 \times 0.2 + 1 \times 0.4 + 2 \times 0.3 + 3 \times 0.3 + 4 \times 0.1 = 2.0$$

and then

$$\mathsf{Var}(X_A) = (1-2)^2 \times 0.3 + (2-2)^2 \times 0.4 + (3-2)^2 \times 0.3 = 0.6,$$
$$\mathsf{Var}(X_B) = (0-2)^2 \times 0.2 + (1-2)^2 \times 0.1 + (2-2)^2 \times 0.3 + (3-2)^2 \times 0.3 + (4-2)^2 \times 0.1 = 1.6.$$

Clearly, the variance of the number of automobiles that are used for official business purposes is greater for company B than for company A.

Variance of a Rondom Variable

Theorem 8

$$\boxed{\mathrm{Var}(X) = E(X^2) - [E(X)]^2} \tag{21}$$

where

$$\boxed{E(X^2) = \sum_{x \in S_X} x^2 P_X(x) \quad \text{if } X \text{ is discrete}} \tag{22}$$

and

$$\boxed{E(X^2) = \int_{-\infty}^{+\infty} x^2 f_X(x)dx \quad \text{if } X \text{ is continuous.}} \tag{23}$$

Variance of a Rondom Variable

Definition 15 (Moment)

For random variable X,

1. The nth moment is $E(X^n)$.
2. The nth central moment is $E[(X - E(X))^n]$.

Variance of a Rondom Variable

Example 26

(a) In Example 6, let X equal the number of bits in error in the next four bits transmitted. Using **(18)**,

$$\mathrm{Var}(X) = (0-0.4)^2 \times 0.6561 + (1-0.4)^2 \times 0.2916 + (2-0.4)^2 \times 0.0486 + (3-0.4)^2 \times 0.0036 + (4-0.4)^2 \times 0.0001 = 0.36.$$

If using **(21)** and Examples 20(a), 22(a), $\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = 0.52 - (0.4)^2 = 0.36.$

(b) In Example 16, X is the current measured in milliamperes, using **(19)** and Example 20(b),

$$\mathrm{Var}(X) = \int_0^{20} (x-10)^2 f_X(x)dx = \int_0^{20} 0.05(x-10)^2 dx = 33.3333.$$

If using **(21)** and Examples 20(b), 22(b), $\mathrm{Var}(X) = E(X^2) - [E(X)]^2 = 133.3333 - (10)^2 = 33.3333.$

Variance and Standard Deviation

✍ Note that $[X - E(X)]^2 \geq 0$. Therefore, its expected value is also nonnegative. That is, for any random variable X

$$\boxed{\mathrm{Var}(X) \geq 0.} \tag{24}$$

The following theorem is related to Theorem 6(2).

Theorem 9

$$\mathrm{Var}(a + bX) = a^2 \mathrm{Var}(X).$$

Variance of a Function of a Rondom Variable

Theorem 10

Let X be a random variable with probability distribution $P_X(x)$ or $f_X(x)$. The variance of the random variable $Y = g(X)$ is

$$\boxed{\mathrm{Var}(Y) = E[g(X) - E(g(X))]^2 = \sum_{x \in S_X} [g(x) - E(g(X))]^2 P_X(x) \quad \text{if } X \text{ is discrete}} \tag{25}$$

and

$$\boxed{\mathrm{Var}(Y) = E[g(X) - E(g(X))]^2 = \int_{-\infty}^{+\infty} [g(x) - E(g(X))]^2 f_X(x)dx \quad \text{if } X \text{ is continuous.}} \tag{26}$$

Variance of a Function of a Rondom Variable

Example 27

Calculate the variance of $g(X) = 2X + 3$, where X is a random variable with probability distribution

X	0	1	2	3
$P(X = x_i)$	1/4	1/8	1/2	1/8

Solution. $\text{Var}(2X + 3) = 4$.

Why?

Variance of a Function of a Rondom Variable

Example 28

Let X be a random variable with density function

$$f_X(x) = \begin{cases} \dfrac{x^2}{3}, & -1 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Find the expected value and the variance of $g(X) = 4X + 3$.

Solution. $\text{Var}(4X + 3) = 51/5$.

Why?

Conditioning a Discrete Random Variable

Definition 16 (Conditional PMF)

Given the event B, with $P(B) > 0$, the conditional probability mass function of X is

$$\boxed{P_{X|B}(x) = P(X = x|B).} \tag{27}$$

Theorem 11

A random variable X resulting from an experiment with event space $B_1, \ldots, B_m$ has PMF

$$\boxed{P_X(x) = \sum_{i=1}^{m} P(B_i) P_{X|B_i}(x).} \tag{28}$$

Conditioning a Discrete Random Variable

Theorem 12

$$\boxed{P_{X|B}(x) = \begin{cases} \dfrac{P_X(x)}{P(B)}, & x \in B, \\ 0, & \text{otherwise.} \end{cases}} \tag{29}$$

Definition 17 (Conditional Expected Value)

The conditional expected value of random variable X given condition B is

$$\boxed{E(X|B) = \sum_{x \in B} x P_{X|B}(x).} \tag{30}$$

Conditioning a Continuous Random Variable

Definition 18 (Conditional PDF given an Event)

For a random variable X with PDF $f_X(x)$ and an event $B \subset S_X$ with $P(B) > 0$, the conditional PDF of X given B is

$$\boxed{f_{X|B}(x) = \begin{cases} \frac{f_X(x)}{P(B)}, & x \in B, \\ 0, & \text{otherwise.} \end{cases}} \tag{31}$$

✍ The function $f_{X|B}(x)$ is a probability model for a new random variable related to X. Thus it has the same properties as any PDF $f_X(x)$.

Theorem 13

Given an event space $\{B_1, B_2, \dots, B_n\}$ and the conditional PDFs $f_{X|B_i}(x)$, $i = 1, 2, \dots, n$,

$$\boxed{f_X(x) = \sum_i P(B_i) f_{X|B_i}(x).} \tag{32}$$

Conditioning a Continuous Random Variable

Definition 19 (Conditional Expected Value)

The conditional expected value of random variable X given condition B is

$$\boxed{E(X|B) = \int_{-\infty}^{+\infty} x f_{X|B}(x)dx.} \quad (33)$$

The conditional expected value of $g(X)$ is

$$\boxed{E(g(X)|B) = \int_{-\infty}^{+\infty} g(x) f_{X|B}(x)dx.} \quad (34)$$

CONTENT

Discrete Uniform Distribution

✍ The simplest discrete random variable is one that assumes only a finite number of possible values, each with equal probability. A random variable X that assumes each of the values $x_1, x_2, \ldots, x_n$, with equal probability $1/n$, is frequently of interest.

Definition 20

A random variable X has a discrete uniform distribution if each of the n values in its range, say $x_1, x_2, \ldots, x_n$, has equal probability. Then,

$$\boxed{P(X = x_i) = \frac{1}{n} \quad \text{for all } i = 1, 2, \ldots, n} \tag{35}$$

Discrete Uniform Distribution

Example 29

The first digit of a part's serial number is equally likely to be any one of the digits 0 through 9. If one part is selected from a large batch and X is the first digit of the serial number, X has a discrete uniform distribution with probability 0.1 for each value in $\{0, 1, 2, \ldots, 9\}$. That is,

$$P(X = x) = 0.1 \quad \text{for } x = 0, 1, \ldots, 9.$$

for each value in R. The probability mass function of X is shown in Fig. 9

Discrete Uniform Distribution

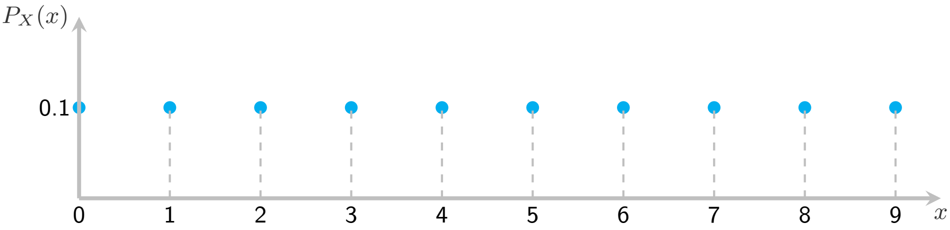


Figure 9: Probability mass function for a discrete uniform random variable.

Discrete Uniform Distribution

Theorem 14

Suppose X is a discrete uniform random variable on the consecutive integers $a, a+1, \ldots, b$, for $a < b$. The mean and the variance of X are

$$\boxed{E(X) = \frac{a+b}{2} \quad \text{and} \quad \mathrm{Var}(X) = \frac{(b-a+1)^2 - 1}{12}.} \tag{36}$$

Example 30

As in Example 2, let the random variable X denote the number of the 48 voice lines that are in use at a particular time. Assume that X is a discrete uniform random variable with a range of 0 to 48. Then,

$$E(X) = \frac{0+48}{2} = 24, \quad \sigma_X = \sqrt{\frac{(48-0+1)^2 - 1}{12}} = 14.14.$$

Practical Interpretation: The average number of lines in use is 24 but the dispersion (as measured by σ) is large. Therefore, at many times far more or fewer than 24 lines are in use.

Bernoulli Distribution

Definition 21 (Bernoulli Random Variable)

X is a Bernoulli random variable if the PMF of X has the form

$$\boxed{P_X(x) = \begin{cases} 1-p, & x = 0, \\ p, & x = 1, \\ 0, & \text{otherwise,} \end{cases}} \tag{37}$$

where the parameter p is in the range $0 < p < 1$.

✍ The probability distribution of this discrete random variable is called the Bernoulli distribution, and is denoted by $X \sim \mathcal{B}(p)$.

Bernoulli Distribution

Example 31

Suppose you test one circuit. With probability 0.1, the circuit is rejected. Let X be the number of rejected circuits in one test. What is $P_X(x)$?

Solution. Because there are only two outcomes in the sample space, $X = 1$ with probability p and $X = 0$ with probability $1 - p$.

$$P_X(x) = \begin{cases} 0.9, & x = 0, \\ 0.1, & x = 1, \\ 0, & \text{otherwise.} \end{cases}$$

Therefore, $X \sim \mathcal{B}(0.1)$.

Bernoulli Distribution

Theorem 15

The mean and variance of the Bernoulli random variable X are

$$\boxed{E(X) = p \quad \text{and} \quad \operatorname{Var}(X) = p(1-p).} \tag{38}$$

Binomial Distribution

Definition 22 (Binomial random variable)

A random experiment consists of n Bernoulli trials such that

1. The trials are independent.
2. Each trial results in only two possible outcomes, labeled as "success" and "failure."
3. The probability of a success in each trial, denoted as p, remains constant

The random variable X that equals the number of trials that result in a success has a binomial random variable with parameters $0 < p < 1$ and $n = 0, 1, 2 \ldots$. The probability mass function of X is

$$\boxed{P_X(x) = \begin{cases} C_n^x p^x (1-p)^{n-x}, & \text{for } x = 0, 1, 2, \ldots, n, \\ 0, & \text{otherwise.} \end{cases}} \tag{39}$$

Binomial Distribution

✍ The probability distribution of this discrete random variable is called the binomial distribution, and is denoted by $X \sim \mathcal{B}(n, p)$ since they depend on the number of trials and the probability of a success on a given trial.

Example 32

Suppose we test 20 circuits and each circuit is rejected with probability 0.1 independent of the results of other tests. Let K equal the number of rejects in the n tests. Find the PMF $P_K(k)$.

Solution. We call each discovery of a defective circuit a success, and each test is an independent trial with success probability 0.2. The event $K = k$ corresponds to k successes in 20 trials, which we have already found, in Equation (39), to be the binomial probability

$$P_K(k) = C_{20}^{k}(0.2)^k 0.8^{20-k}, \quad \text{for } k = 0, 1, 2, \ldots, 20.$$

K is an example of a binomial random variable and $K \sim \mathcal{B}(20, 0.2)$.

Binomial Distribution

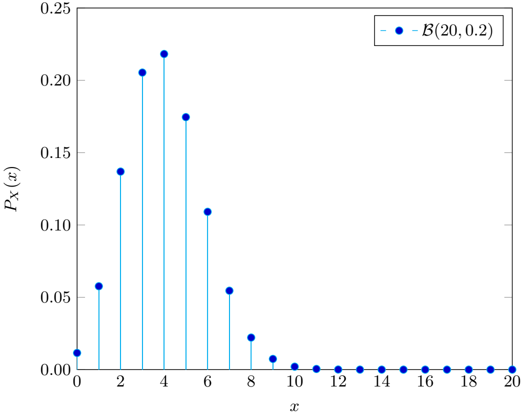


Figure 10: Binomial distributions for $n = 20$ and $p = 0.2$

Binomial Distribution

Theorem 16

The mean and variance of the binomial random variable X are

$$\boxed{E(X) = np \quad \text{and} \quad \text{Var}(X) = np(1-p).} \tag{40}$$

Theorem 17

If X is a binomial random variable with parameters p and n,

$$\boxed{(n+1)p - 1 \leq \text{mod}(X) \leq (n+1)p.} \tag{41}$$

Binomial Distribution

Example 33

For the number of transmitted bits received in error in Example 6, $n = 4$ and $p = 0.1$, so

$$E(X) = np = 4 \times 0.1 = 0.4 \quad \text{and} \quad \mathrm{Var}(X) = np(1-p) = 4 \times 0.1 \times 0.9 = 0.36$$

and these results match those obtained from a direct calculation in Examples 20(a) and 26(a).

Poisson Distributionn

Definition 23 (Poisson random variable)

X is a Poisson random variable if the PMF of X has the form

$$\boxed{P_X(x) = \begin{cases} \dfrac{\lambda^x e^{-\lambda}}{x!}, & x = 0, 1, 2, \ldots \\ 0, & \text{otherwise}, \end{cases}} \tag{42}$$

where the parameter λ is in the range $\lambda > 0$.

✍ The symbol $e \simeq 2.71828\ldots$ is evaluated using your scientific calculator, which should have a function such as e^x. Probability distribution of Poisson random variable is called the Poisson distribution, and is denoted by $X \sim \mathcal{P}(\lambda)$.

Poisson Distributionn

Remark 2

Here are some examples of experiments for which the random variable X can be modeled by the Poisson random variable:

1. The number of calls received by a switchboard during a given period of time.
2. The number of bacteria per small volume of fluid.
3. The number of customer arrivals at a checkout counter during a given minute.
4. The number of machine breakdowns during a given day.
5. The number of traffic accidents at a given intersection during a given time period.

Poisson Distributionn

Example 34

The number of hits at a Web site in any time interval is a Poisson random variable. A particular site has on average $\alpha = 2$ hits per second.

(a) What is the probability that there are no hits in an interval of 0.25 seconds?

(b) What is the probability that there are no more than two hits in an interval of one second?

Solution. (a) In an interval of 0.25 seconds, the number of hits H is a Poisson random variable with $\lambda = \alpha T = (2\text{hits}/s) \times (0.25s) = 0.5$ hits. The PMF of H is

$$P_H(h) = \begin{cases} \dfrac{(0.5)^h \times e^{-5}}{h!}, & h = 0, 1, 2, \ldots \\ 0, & \text{otherwise.} \end{cases}$$

The probability of no hits is

$$P(H = 0) = P_H(0) = \frac{(0.5)^0 \times e^{-0.5}}{0!} = 0.607.$$

Poisson Distributionn

Solution. (b) In an interval of 1 second, $\lambda = \alpha T = (2\text{hits}/s) \times (1s) = 2$ hits. Letting J denote the number of hits in one second, the PMF of J is

$$P_J(j) = \begin{cases} \dfrac{(2)^j \times e^{-2}}{j!}, & j = 0, 1, 2, \ldots \\ 0, & \text{otherwise.} \end{cases}$$

To find the probability of no more than two hits, we note that $\{J \le 2\} = \{J = 0\} \cup \{J = 1\} \cup \{J = 2\}$ is the union of three mutually exclusive events. Therefore,

$$\begin{aligned} P(J \le 2) &= P(J = 0) + P(J = 1) + P(J = 2) = P_J(0) + P_J(1) + P_J(2) \\ &= e^{-2} + \frac{2^1 \times e^{-2}}{1!} + \frac{2^2 \times e^{-2}}{2!} = 0.677. \end{aligned}$$

Poisson Distributionn

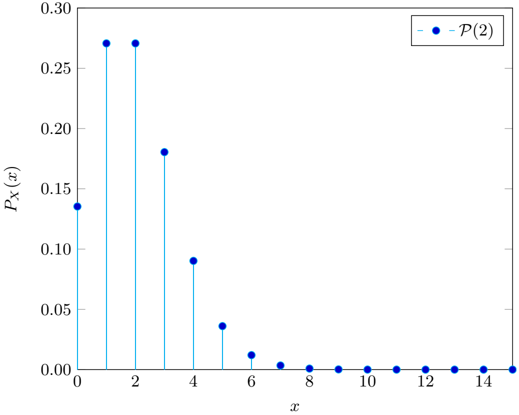


Figure 11: Poisson distributions for $\lambda = 2$

Poisson Distribution

Theorem 18

If X is a Poisson random variable with parameter λ, then

$$\boxed{E(X) = \lambda \quad \text{and} \quad \mathrm{Var}(X) = \lambda} \tag{43}$$

Approximation of Binomial Distribution by a Poisson Distribution

Theorem 19

Let X be a binomial random variable with probability distribution $\mathcal{B}(n,p)$. When $n \to \infty$, $p \to 0$, and $np \to \mu$ as $n \to \infty$ remains constant,

$$\boxed{\mathcal{B}(n,p) \to \mathcal{P}(\lambda) \quad \text{as} \quad n \to \infty.} \tag{44}$$

Remark 3

The Poisson distribution provides a simple, easy-to-compute, and accurate approximation to binomial probabilities when n is large and $\lambda = np$ is small, preferably with $np < 7$.

Approximation of Binomial Distribution by a Poisson Distribution

Example 35

Suppose a life insurance company insures the lives of 5000 men aged 42. If actuarial studies show the probability that any 42-year-old man will die in a given year to be 0.001, find the exact probability that the company will have to pay $X = 4$ claims during a given year.

Solution. The exact probability is given by the binomial distribution as

$$P(X = 4) = \frac{5000!}{4!4996!}(0.001)^4(0.999)^{4996}$$

for which binomial tables are not available. To compute $P(X = 4)$ without the aid of a computer would be very time-consuming, but the Poisson distribution can be used to provide a good approximation to $P(X = 4)$. Computing $\lambda = np = (5000)(0.001) = 5$ and substituting into the formula for the Poisson probability distribution, we have

$$P(X = 4) \simeq \frac{5^4}{4!}e^{-5} = \frac{(625)(0.006738)}{24} = 0.175.$$

CONTENT

Continuous Uniform Distribution

Definition 24 (Uniform random variable)

X is a uniform $\mathcal{U}[a,b]$ random variable if the PDF of X is

$$\boxed{f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b, \\ 0, & \text{otherwise}, \end{cases}} \tag{45}$$

where the two parameters are $b > a$.

✍ The probability distribution of this continuous random variable is called the uniform distribution, and is denoted by $X \sim \mathcal{U}[a,b]$.

Continuous Uniform Distribution

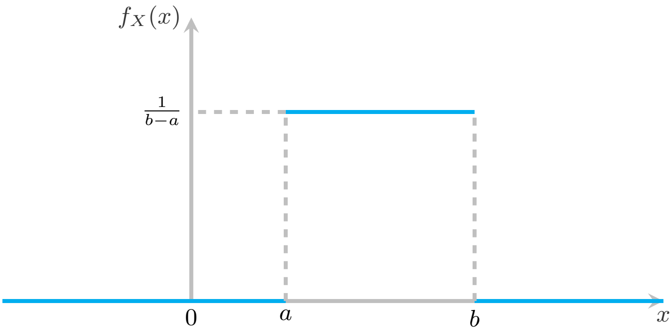


Figure 12: The density function for a random variable on the interval $[a, b]$

Continuous Uniform Distribution

Example 36

Suppose that a large conference room at a certain company can be reserved for no more than 4 hours. Both long and short conferences occur quite often. In fact, it can be assumed that the length X of a conference has a uniform distribution on the interval $[0, 4]$. (a) What is the probability density function? (b) What is the probability that any given conference lasts at least 3 hours?

Solution.

(a) The appropriate density function for the uniformly distributed random variable X in this situation is

$$f(x) = \begin{cases} \frac{1}{4}, & 0 \le x \le 4, \\ 0, & \text{otherwise.} \end{cases}$$

(b) $P(X \ge 3) = \int_3^4 \frac{1}{4} dx = \frac{1}{4}.$

Continuous Uniform Distribution

Theorem 20

If X is a uniform $\mathcal{U}[a,b]$ random variable,

1. The CDF of X is

$$\boxed{F_X(x) = \begin{cases} 0, & x \le a, \\ \dfrac{x-a}{b-a}, & a < x \le b, \\ 1, & x > b. \end{cases}} \tag{46}$$

2. The expected value and the variance of X are

$$\boxed{E(X) = \frac{a+b}{2} \quad \text{and} \quad \mathrm{Var}(X) = \frac{(b-a)^2}{12}.} \tag{47}$$

Continuous Uniform Distribution

✍ If X is a uniform $\mathcal{U}[a,b]$ random variable,

$$\boxed{P(\alpha < X < \beta) = \frac{\beta-\alpha}{b-a}, \quad \alpha, \beta \in \mathbb{R}.} \tag{48}$$

Example 37

The random variable X in Example 16 is a uniform $\mathcal{U}[0,20]$ random variable. Then,

$$E(X) = \frac{0+20}{2} = 10 \text{ mA} \quad \text{and} \quad \text{Var}(X) = \frac{20^2}{12} = 33.3333 \text{ mA}^2$$

and these results match those obtained from a direct calculation in Examples 20(b) and 22(b).

Exponential Distribution

Definition 25 (Exponential random variable)

The random variable X that equals the distance between successive events of a Poisson process with mean number of events $\lambda > 0$ per unit interval is an exponential random variable with parameter λ. The probability density function of X is

$$\boxed{f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & \text{otherwise.} \end{cases}} \tag{49}$$

✍ The probability distribution of this continuous random variable is called the exponential distribution, and is denoted by $X \sim \mathsf{Exp}(\lambda)$.

Exponential Distribution

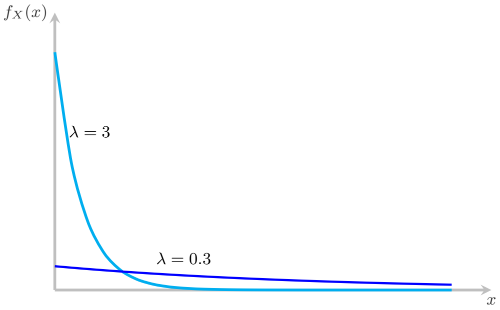


Figure 13: Probability density function of exponential random variables for selected values of λ

Exponential Distribution

Example 38

The probability that a telephone call lasts no more than t minutes is often modeled as an exponential CDF.

$$F_T(t) = \begin{cases} 1 - e^{-t/3}, & t \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

(a) What is the PDF of the duration in minutes of a telephone conversation? (b) What is the probability that a conversation will last between 2 and 4 minutes?

Exponential Distribution

Solution.

(a) We find the PDF of T by taking the derivative of the CDF:

$$f_T(t) = \frac{dF_T(t)}{dt} = \begin{cases} \frac{1}{3}e^{-t/3}, & t \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

Therefore, observing Definition 25, we recognize that T is an exponential $(\lambda = 1/3)$ random variable.

(b) The probability that a call lasts between 2 and 4 minutes is

$$P(2 \leq T \leq 4) = F_T(4) - F_T(2) = e^{-2/3} - e^{-4/3} = 0.250.$$

Exponential Distribution

Theorem 21

If X is an exponential $\text{Exp}(\lambda)$ random variable,

(a) $F_X(x) = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0, \\ 0, & \text{otherwise.} \end{cases}$

(b) $E(X) = 1/\lambda$.

(c) $\text{Var}(X) = 1/\lambda^2$.

(d) $P(X > x) = e^{-\lambda x} \quad x \geq 0$.

Example 39

In Example 38, (a) what is $E(T)$, the expected duration of a telephone call? (b) What are the variance and standard deviation of T?

Solution. Using Theorem 21, (a) $E(T) = \frac{1}{1/3} = 3$ minutes.

(b) $\text{Var}(T) = \frac{1}{1/9} = 9$ and the standard deviation is $\sigma_T = \sqrt{\text{Var}(T)} = 3$ minutes.

Exponential Distribution

Example 40

Let X denote the time between detections of a particle with a Geiger counter and assume that X has an exponential distribution with $E(X) = 1.4$ minutes. The probability that we detect a particle within 30 seconds of starting the counter is

$$P(X < 0.5 \text{ minute}) = F_X(0.5) - F_X(0) = 1 - e^{-0.5/1.4} = 0.3.$$

In this calculation, all units are converted to minutes.
Now, suppose we turn on the Geiger counter and wait 3 minutes without detecting a particle. What is the probability that a particle is detected in the next 30 seconds? The requested probability can be expressed as the conditional probability that $P(X < 3.5|X > 3)$. From the definition of conditional probability,

$$P(X < 3.5|X > 3) = \frac{P(3 < X < 3.5)}{P(X > 3)} = \frac{F_X(3.5) - F_X(3)}{1 - F_X(3)} = \frac{0.035}{0.117} = 0.30.$$

Exponential Distribution

✍ Example 40 illustrates the lack of memory property of an exponential random variable, and a general statement of the property follows. In fact, the exponential distribution is the only continuous distribution with this property.

$$\boxed{P(X > t + s | X > t) = P(X > s).} \tag{50}$$

Normal Distribution

✍ The most important continuous probability distribution in the entire field of statistics is the normal distribution. Its graph, called the normal curve, is the bell-shaped curve of Figure 14, which approximately describes many phenomena that occur in nature, industry, and research.

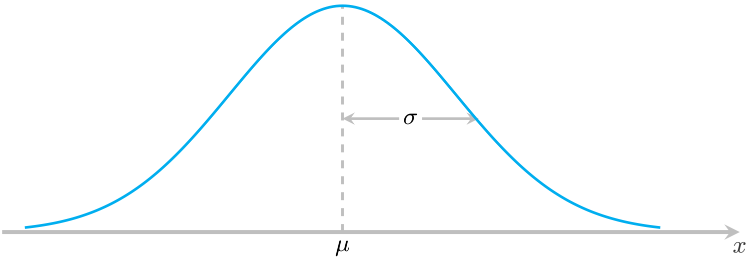


Figure 14: The normal curve

Normal Distribution

✍ A continuous random variable X having the bell-shaped distribution of Figure 14 is called a normal random variable. The mathematical equation for the probability distribution of the normal variable depends on the two parameters μ and σ, its mean and standard deviation, respectively. We denote the normal distribution by $\mathcal{N}(\mu, \sigma^2)$.

Definition 26

The PDF of the normal random variable X, with mean μ and variance σ^2, is

$$\boxed{f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\dfrac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty < x < \infty,} \tag{51}$$

the symbols e and π are mathematical constants given approximately by 2.7183 and 3.1416, respectively.

Normal Distribution

✍ In Figure 15, we have sketched two normal curves having the same standard deviation but different means. The two curves are identical in form but are centered at different positions along the horizontal axis.

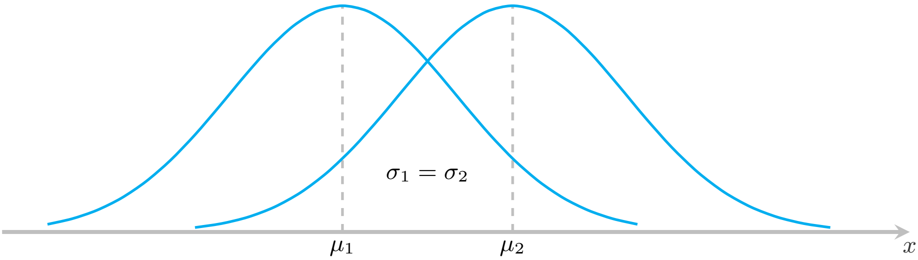


Figure 15: Normal curves with $\mu_1 < \mu_2$ and $\sigma_1 = \sigma_2$

Normal Distribution

✍ In Figure 16, we have sketched two normal curves with the same mean but different standard deviations.

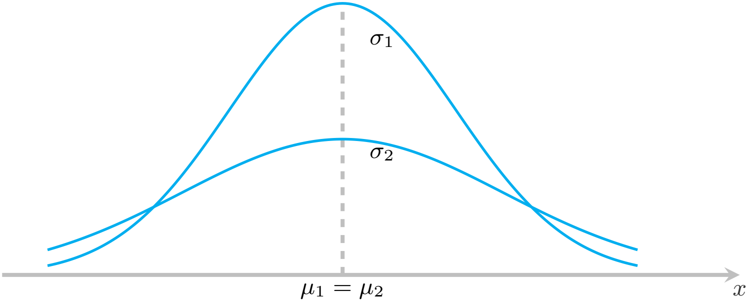


Figure 16: Normal curves with $\mu_1 = \mu_2$ and $\sigma_1 < \sigma_2$

Normal Distribution

✍ Figure 17 shows two normal curves having different means and different standard deviations.

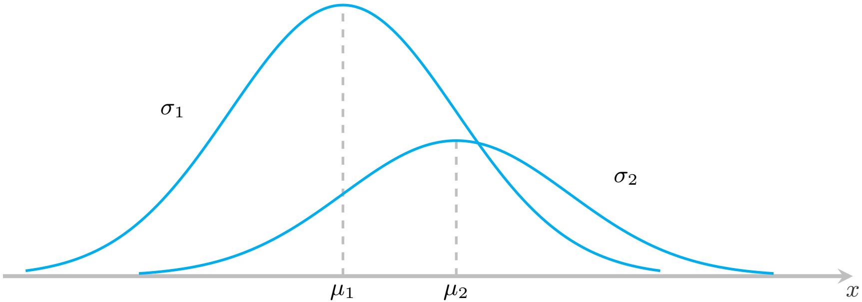


Figure 17: Normal curves with $\mu_1 < \mu_2$ and $\sigma_1 < \sigma_2$

Normal Distribution

Theorem 22

The mean and variance of the normal random variable are μ and σ^2, respectively. Hence, the standard deviation is σ.

Theorem 23

If X is normal random variable $\mathcal{N}(\mu, \sigma^2)$, $Y = aX + b$ is normal random variable $\mathcal{N}(a\mu + b, a^2\sigma^2)$.

Standard Normal Distribution. Area Under the Normal Curve

✍ Put

$$Z = \frac{X - \mu}{\sigma}.$$

If X is a normal random variable with $E(X) = \mu$ and $\text{Var}(X) = \sigma^2$ then Z is seen to be a normal random variable with $E(Z) = 0$ and $\text{Var}(Z) = 1$.

Definition 27 (Standard normal distribution)

The distribution of a normal random variable with mean 0 and variance 1 is called a standard normal distribution.

Standard Normal Distribution. Area Under the Normal Curve

Definition 28 (PDF and CDF)

The PDF of the standard normal random variable Z is

$$\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}.$$

The CDF of the standard normal random variable Z is

$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int\limits_{-\infty}^{z} e^{-\frac{u^2}{2}} du.$$

Standard Normal Distribution. Area Under the Normal Curve

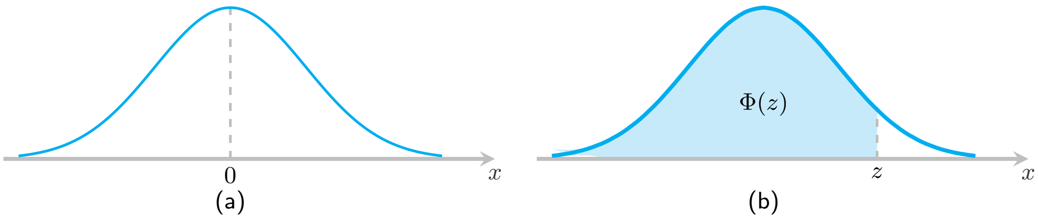


Figure 18: (a) Standardizing a normal random variable; (b) $\Phi(z) = P(Z < z)$

Standard Normal Distribution. Area Under the Normal Curve

✍ Given a table of values of $\Phi(z)$, Table 1, we use the following theorem to find probabilities of a normal random variable with parameters μ and σ^2.

Theorem 24

If X is a normal random variable $\mathcal{N}(\mu, \sigma^2)$, the CDF of X is

$$F_X(x) = \Phi\left(\frac{x-\mu}{\sigma}\right).$$

The probability that X is in the interval (x_1, x_2) is

$$P(x_1 < X < x_2) = \Phi\left(\frac{x_2-\mu}{\sigma}\right) - \Phi\left(\frac{x_1-\mu}{\sigma}\right).$$

Standard Normal Distribution. Area Under the Normal Curve

Corollary 2

(a) $P(X < x_2) = \Phi\left(\dfrac{x_2 - \mu}{\sigma}\right)$.

(b) $P(X > x_1) = 1 - \Phi\left(\dfrac{x_1 - \mu}{\sigma}\right)$.

(c) $P(|X - \mu| < \varepsilon) = 2\Phi\left(\frac{\varepsilon}{\sigma}\right) - 1$.

Standard Normal Distribution. Area Under the Normal Curve

✍ The curve of any continuous probability distribution or probability density function is constructed so that the area under the curve bounded by the two ordinates $x = x_1$ and $x = x_2$ equals the probability that the random variable X assumes a value between $x = x_1$ and $x = x_2$. Thus, for the normal curve in Figure 19,

$$P(x_1 < X < x_2) = \frac{1}{\sigma\sqrt{2\pi}} \int\limits_{x_1}^{x_2} e^{-\frac{(x-\mu)^2}{2\sigma^2}}\, dx$$

is represented by the area of the shaded region.

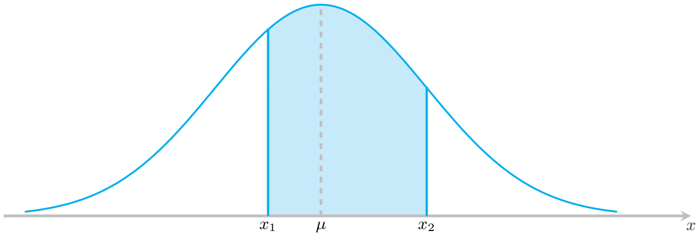


Figure 19: $P(x_1 < X < x_2) =$ area of the shaded region

Standard Normal Distribution. Area Under the Normal Curve

Remark 4

(a) In using Theorem 24, we transform values of a norm random variable, X, to equivalent values of the standard normal random variable, Z. For a sample value x of the random variable X, the corresponding sample value of Z is

$$z = \frac{x - \mu}{\sigma} \quad \text{or equivalently,} \quad x = \mu + z\sigma. \tag{52}$$

The original and transformed distributions are illustrated in Figure 20. Since all the values of X falling between x_1 and x_2 have corresponding z values between z_1 and z_2, the area under the X-curve between the ordinates $x = x_1$ and $x = x_2$ in Figure 20 equals the area under the Z-curve between the transformed ordinates $z = z_1$ and $z = z_2$.

(b) The probability distribution for Z, shown in Figure 18(b), is called the standardized normal distribution because its mean is 0 and its standard deviation is 1. Values of Z on the left side of the curve are negative, while values on the right side are positive. The area under the standard normal curve to the left of a specified value of Z say, z is the probability $P(Z < z)$. This cumulative area is recorded in Tables 1-2 and is shown as the shaded area in Figure 18(b).

Standard Normal Distribution. Area Under the Normal Curve

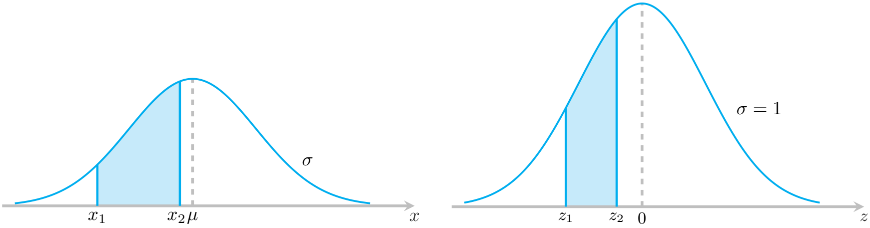


Figure 20: The original and transformed normal distributions

Standard Normal Distribution. Area Under the Normal Curve

z	0	1	2	3	4	5	6	7	8	9
0.0	0.50000	50399	50798	51197	51595	51994	52392	52790	53188	53586
0.1	53983	54380	54776	55172	55567	55962	56356	56749	57142	57535
0.2	57926	58317	58706	59095	59483	59871	60257	60642	61026	61409
0.3	61791	62172	62556	62930	63307	63683	64058	64431	64803	65173
0.4	65542	65910	66276	66640	67003	67364	67724	68082	68439	68739
0.5	69146	69447	69847	70194	70544	70884	71226	71566	71904	72240
0.6	72575	72907	73237	73565	73891	74215	74537	74857	75175	75490
0.7	75804	76115	76424	76730	77035	77337	77637	77935	78230	78524
0.8	78814	79103	79389	79673	79955	80234	80511	80785	81057	81327
0.9	81594	81859	82121	82381	82639	82894	83147	83398	83646	83891
1,0	**84134**	84375	84614	84850	85083	85314	85543	85769	85993	86214
1.1	86433	86650	86864	87076	87286	87493	87698	87900	88100	88298
1.2	88493	88686	88877	89065	89251	89435	89617	89796	89973	90147
1.3	90320	90490	90658	90824	90988	91149	91309	91466	91621	91774
1.4	91924	92073	92220	92364	92507	92647	92786	92922	93056	93189
1.5	**93319**	93448	93574	93699	93822	93943	94062	94179	94295	94408
1.6	94520	94630	94738	94845	94950	95053	95154	95254	95352	95449
1.7	95543	95637	95728	95818	95907	95994	96080	96164	96246	96327
1.8	96407	96485	96562	96638	96712	96784	96856	96926	96995	97062
1.9	97128	97193	97257	97320	97381	97441	97500	97558	97615	97670

Table 1: The values of $\Phi(Z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{Z} e^{\frac{-t^2}{2}} dt$

Standard Normal Distribution. Area Under the Normal Curve

z	0	1	2	3	4	5	6	7	8	9
2.0	**97725**	97778	97831	97882	97932	97982	98030	98077	98124	98169
2.1	98214	98257	98300	98341	98382	98422	99461	98500	98537	98574
2.2	98610	98645	98679	98713	98745	98778	98809	98840	98870	98899
2.3	98928	98956	98983	99010	99036	99061	99086	99111	99134	99158
2.4	99180	99202	99224	99245	99266	99285	99305	99324	99343	99361
2.5	99379	99396	99413	99430	99446	99261	99477	99492	99506	99520
2.6	99534	99547	99560	99573	99585	99598	99609	99621	99632	99643
2.7	99653	99664	99674	99683	99693	99702	99711	99720	99728	99763
2.8	99744	99752	99760	99767	99774	99781	99788	99795	99801	99807
2.9	99813	99819	99825	99831	99836	99841	99846	99851	99856	99861
3.0	0.99865	3.1	99903	3.2	99931	3.3	99952	3.4	99966	
3.5	99977	3.6	99984	3.7	99989	3.8	99993	3.9	99995	
4.0	999968									
4.5	999997									
5.0	99999997									

Table 2: The values of $\Phi(Z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{Z} e^{\frac{-t^2}{2}} dt$

Standard Normal Distribution. Area Under the Normal Curve

Example 41

Suppose the current measurements in a strip of wire are assumed to follow a normal distribution with a mean of 10 milliamperes and a variance of 4 (milliamperes)^2. What is the probability that a measurement will exceed 13 milliamperes?

Solution. Let X denote the current in milliamperes. The requested probability can be represented as $P(X > 13)$. Using Corollary 2 and Table 1,

$$P(X > 13) = 1 - \Phi\left(\frac{13 - 10}{2}\right) = 1 - \Phi(1.5) = 1 - 0.93319 = 0.06681.$$

Standard Normal Distribution. Area Under the Normal Curve

Remark 1

To find probabilities of norm random variables, we use the values of (z) presented in Tables 1-2. Note that this table contains entries only for $z \geq 0$. For negative values of z, we apply the following property of $\Phi(z)$.

Theorem 25

$$\Phi(-z) = 1 - \Phi(z).$$

Standard Normal Distribution. Area Under the Normal Curve

✍ Figure 21 displays the symmetry properties of $\Phi(z)$. Both graphs contain the standard normal PDF. In Figure 21(a), the shaded area under the PDF is $\Phi(z)$. Since the area under the PDF equals 1, the unshaded area the PDF is $1 - \Phi(z)$. In Figure 21(b), the shaded area on the right is $1 - \Phi(z)$ and the shaded area on the left is $\Phi(-z)$. This graph demonstrates that $\Phi(-z) = 1 - \Phi(z)$.

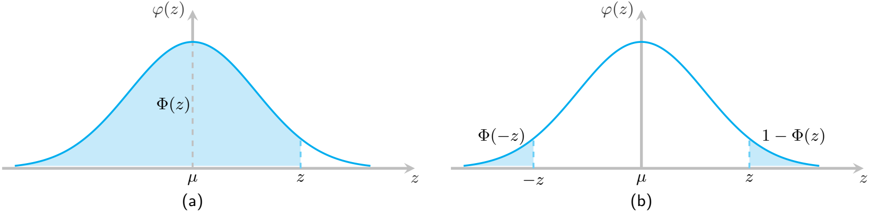


Figure 21: Standardized normal distribution

Standard Normal Distribution. Area Under the Normal Curve

Example 42

If X is the norm random variable $\mathcal{N}(61, 10^2)$, (a) what is $P(X \leq 46)$? (b) What is $P(51 < X < 71)$?

Solution.

(a) Applying Corollary 2(a), Theorem 25, we have

$$P(X \leq 46) = \Phi\left(\frac{46-61}{10}\right) = \Phi(-1.5) = 1 - \Phi(1.5) = 1 - 0.93319 = 0.06681.$$

(b) Applying Equation (52), we find that the event $\{51 < X < 71\}$ corresponds to $\{-1 < Z < 1\}$. The probability of this event is

$$\Phi(1) - \Phi(-1) = \Phi(1) - [1 - \Phi(1)] = 2\Phi(1) - 1 = 0.68268.$$

Standard Normal Distribution. Area Under the Normal Curve

✍ Some useful results concerning a normal distribution are summarized below and in Fig. 22. For any normal random variable,

$$P(\mu - \sigma < X < \mu + \sigma) = 0.6827,$$
$$P(\mu - 2\sigma < X < \mu + 2\sigma) = 0.9545,$$
$$P(\mu - 3\sigma < X < \mu + 3\sigma) = 0.9973.$$

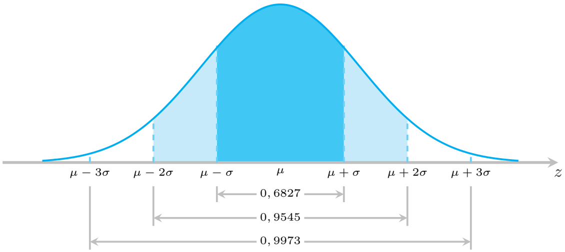


Figure 22: Probabilities associated with a normal distribution

The Normal Approximation to the Binomial Probability Distribution

✍ Probabilities associated with binomial experiments are readily obtainable from the formula $\mathcal{B}(n, p)$ of the binomial distribution when n is small. In the previous section, we illustrated how the Poisson distribution can be used to approximate binomial probabilities when n is quite large and p is very close to 0 or 1. Both the binomial and the Poisson distributions are discrete. We now state a theorem that allows us to use areas under the normal curve to approximate binomial properties when n is sufficiently large.

Theorem 26

If X is a binomial random variable with mean $\mu = np$ and variance $\sigma^2 = np(1-p)$, then the limiting form of the distribution of

$$Z = \frac{X - np}{\sqrt{np(1-p)}},$$

as $n \to \infty$, is the standard normal distribution $\mathcal{N}(0, 1)$.

The Normal Approximation to the Binomial Probability Distribution

✍ To illustrate the normal approximation to the binomial distribution, we first draw the histogram for $\mathcal{B}(15, 0.4)$ and then superimpose the particular normal curve having the same mean and variance as the binomial variable X. Hence, we draw a normal curve with $\mu = np = (15)(0.4) = 6$ and $\sigma^2 = np(1-p) = (15)(0.4)(0.6) = 3.6$. Scatter plot of $\mathcal{B}(15, 0.4)$ and the corresponding superimposed normal curve, which is completely determined by its mean and variance, are illustrated in Figure 23.

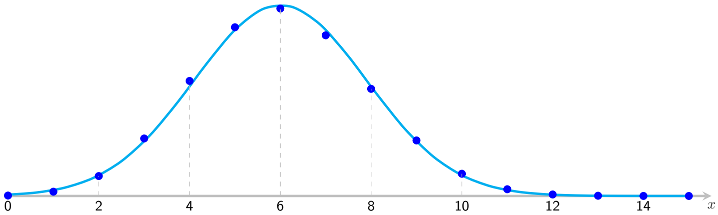


Figure 23: Normal approximation of $\mathcal{B}(15, 0.4)$

The Normal Approximation to the Binomial Probability Distribution

Definition 29 (Normal Approximation to the Binomial Distribution)

Let X be a binomial random variable with n trials and probability p of success. The probability distribution of X is approximated using a normal curve with

$$\mu = np \quad \text{and} \quad \sigma = \sqrt{np(1-p)},$$

and

$$P(X = k) \simeq \Phi\left(\frac{k + 0.5 - \mu}{\sigma}\right) - \Phi\left(\frac{k - 0.5 - \mu}{\sigma}\right) \tag{53}$$

and

$$P(k_1 \leq X \leq k_2) \simeq \Phi\left(\frac{k_2 + 0,5 - \mu}{\sigma}\right) - \Phi\left(\frac{k_1 - 0,5 - \mu}{\sigma}\right) \tag{54}$$

and the approximation will be good if np and $n(1-p)$ are greater than or equal to 5.

The Normal Approximation to the Binomial Probability Distribution

Example 43

Use the normal curve to approximate the probability that $X = 8$, 9, or 10 for a binomial random variable with $n = 25$ and $p = 0.5$. Compare this approximation to the exact binomial probability.

Solution. You can find the exact binomial probability for this example because there are cumulative binomial tables for $n = 25$,

$$P(X = 8) + P(X = 9) + P(X = 10) = \left(C_{25}^{8} + C_{25}^{9} + C_{25}^{10}\right)(0.5)^{25} \simeq 0.190535.$$

To use the normal approximation, first find the appropriate mean and standard deviation for the normal curve: $\mu = np = 12.5$, $\sigma = \sqrt{np(1-p)} = 2.5$. It follows from (54) that

$$P(8 \leq X \leq 10) = \Phi(-0.8) - \Phi(-2) = 0.18911.$$

You can compare the approximation, 0.18911, to the actual probability, 0.190535. They are quite close!

The Normal Approximation to the Binomial Probability Distribution

Remark 5

The normal approximation to the binomial probabilities will be adequate if both $np \geq 5$ and $n(1-p) \geq 5$.

Example 44

The reliability of an electrical fuse is the probability that a fuse, chosen at random from production, will function under its designed conditions. A random sample of 1000 fuses was tested and $X = 27$ defectives were observed. Calculate the approximate probability of observing 27 or more defectives, assuming that the fuse reliability is 0.98.

The Normal Approximation to the Binomial Probability Distribution

Solution. The probability of observing a defective when a single fuse is tested is $p = 0.02$, given that the fuse reliability is 0.98. Then $\mu = np = 20$, $\sigma = \sqrt{np(1-p)} = 4.43$.
The probability of 27 or more defective fuses, given $n = 1000$, is

$$P(X \geq 27) = P(X = 27) + P(X = 28) + \cdots + P(X = 1000).$$

It is appropriate to use the normal approximation to the binomial probability because $np = 20$ and $n(1-p) = 980$ are both greater than 5. So

$$\begin{aligned} P(27 \leq X \leq 1000) &= \Phi\left(\frac{1000 + 0.5 - 20}{4.43}\right) - \Phi\left(\frac{27 - 0.5 - 20}{4.43}\right) \\ &= 1 - \Phi(1.47) = 1 - 0.92922 = 0.07078. \end{aligned}$$

CONTENT

Discrete Random Variables

Exercise 1

The random variable N has PMF

$$P_N(n) = \begin{cases} c(1/2)^n, & n = 0, 1, 2, \\ 0, & \text{otherwise.} \end{cases}$$

(a) What is the value of the constant c?

(b) What is $P(N \leq 1)$?

(c) What is the probability distribution of X?

Solution. (a) $4/7$ (b) $6/7$

Discrete Random Variables

Exercise 2

The random variable V has PMF

$$P_V(v) = \begin{cases} cv^2, & v = 1,2,3,4, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the value of the constant c.

(b) Find $P(V \in \ u^2 | u = 1, 2, 3, \dots)$.

(c) Find the probability that V is an even number.

(d) Find $P(V > 2)$.

Solution. (a) $c = 1/30$ (b) $7/30$ (c) $2/3$ (d) $5/6$

Discrete Random Variables

Exercise 3

Suppose when a baseball player gets a hit, a single is twice as likely as a double which is twice as likely as a triple which is twice as likely as a home run. Also, the player's batting average, i.e., the probability the player gets a hit, is 0.300. Let B denote the number of bases touched safely during an at-bat. For example, $B = 0$ when the player makes an out, $B = 1$ on a single, and so on.

(a) What is the PMF of B?

(b) Find and sketch the CDF of B.

Solution. (a) $$P_B(b) = \begin{cases} 0.7, & b = 0 \\ 0.16, & b = 1 \\ 0.08, & b = 2 \\ 0.04, & b = 3 \\ 0.02, & b = 4 \\ 0, & \text{otherwise.} \end{cases}$$

Discrete Random Variables

Exercise 4

In a package of M&Ms, Y, the number of yellow M&Ms, is uniformly distributed between 5 and 15.

(a) What is the PMF of Y?

(b) What is $P(Y < 10)$?

(c) What is $P(Y > 12)$?

(d) What is $P(8 \leq X \leq 12)$?

Solution. (b) $5/11$ (c) $3/11$ (d) $5/11$

Discrete Random Variables

Exercise 5

A civil engineer is studying a left-turn lane that is long enough to hold 7 cars. Let X be the number of cars in the lane at the end of a randomly chosen red light. The engineer believes that the problem that $(X = x)$ is proportional to $(x+1)(8-x)$ for $x = 0, 1, \dots, 7$.

(a) Find the probability mass function of X.

(b) Find the probability that X will be at least 5.

Solution. (a) $c = 1/200$ (b) $p = 1/3$

Discrete Random Variables

Exercise 6

A midterm test has 4 multiple choice questions with four choices with one correct answer each. If you just randomly guess on each of the 4 questions, what is the probability that you get exactly 2 questions correct? Assume that you answer all and you will get $(+5)$ points for 1 question correct, (-2) points for 1 question wrong. Let X is number of points that you get. Find the probability mass function of X and the expected value of X.

Discrete Random Variables

Exercise 7

When a conventional paging system transmits a message, the probability that the message will be received by the pager it is sent to is p. To be confident that a message is received at least once, a system transmits the message n times.

(a) Assuming all transmissions are independent, what is the PMF of K, the number of times the pager receives the same message?

(b) Assume $p = 0.8$. What is the minimum value of n that produces a probability of 0.95 of receiving the message at least once?

Solution. (a) $P_K(k) = \begin{cases} C_n^k p^k (1-p)^{n-k}, & k = 0, 1, \ldots, n \\ 0, & \text{otherwise.} \end{cases}$ (b) $n = 2$

Discrete Random Variables

Exercise 8

When a two-way paging system transmits a message, the probability that the message will be received by the pager it is sent to is p. When the pager receives the message, it transmits an acknowledgment signal (ACK) to the paging system. If the paging system does not receive the ACK, it sends the message again.

(a) What is the PMF of N, the number of times the system sends the same message?

(b) The paging company wants to limit the number of times it has to send the same message. It has a goal of $P(N \leq 3) \geq 0.95$. What is the minimum value of p necessary to achieve the goal?

Solution. (a) $P_N(n) = \begin{cases} (1-p)^{n-1}p, & k = 1, 2, \ldots \\ 0, & \text{otherwise.} \end{cases}$ (b) $p \geq 0.6316$

Discrete Random Variables

Exercise 9

The random variable X has CDF

$$F_X(x) = \begin{cases} 0, & x \le -1, \\ 0.2, & -1 < x \le 0, \\ 0.7, & 0 < x \le 1, \\ 1, & x > 1. \end{cases}$$

(a) Draw a graph of the CDF.

(b) Write $P_X(x)$, the PMF of X. Be sure to write the value of $P_X(x)$ for all x from $-\infty$ to ∞.

(c) Write the probability distribution of X.

(d) Find the expected value of the random variable X.

(e) Let $V = g(X) = |X|$. Find $P_V(v)$, $F_V(v)$, $E(V)$.

(f) Find the variance of the random variable X.

Discrete Random Variables

Exercise 10

The random variable X has CDF

$$F_X(x) = \begin{cases} 0, & x \le -3, \\ 0.4, & -3 < x \le 5, \\ 0.8, & 5 < x \le 7, \\ 1, & x > 7. \end{cases}$$

(a) Draw a graph of the CDF.

(b) Write $P_X(x)$, the PMF of X.

(c) Write the probability distribution of X.

(d) Find the expected value of the random variable X

Discrete Random Variables

Exercise 11

Let X have the uniform PMF

$$P_X(x) = \begin{cases} 0.01, & x = 1, 2, \ldots, 100, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find a mode x_{mod} of X. If the mode is not unique, find the set X_{mod} of all modes of X.

(b) Find a median x_{med} of X. If the median is not unique, find the set X_{med} of all numbers x that are medians of X.

Discrete Random Variables

Exercise 12

In an experiment to monitor two calls, the PMF of N, the number of voice calls, is

$$P_N(n) = \begin{cases} 0.2, & n = 0, \\ 0.7, & n = 1, \\ 0.1, & n = 2, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find $E(N)$, the expected number of voice calls.

(b) Find $E(N^2)$, the second moment of N.

(c) Find $\mathrm{Var}(N)$, the variance of N.

(d) Find $\sigma(N)$, the standard deviation of N.

Solution. (a) 0.9 (b) 1.1 (c) 0.29 (d) $\sqrt{0.29}$.

Discrete Random Variables

Exercise 13

A box contains 15 ping-pong balls, 10 of which are new. The first time we draw 3 balls to play, then return them to the box. The second time, we take out 3 balls. Let X be the number of new balls out of 3 balls drawn in the second time. Find the expected value of the random variable X.

Solution. 1.6

Continuous Random Variables

Exercise 14

The cumulative distribution function of random variable X is

$$F_X(x) = \begin{cases} 0, & x \le -1, \\ (x+1)/2, & -1 < x \le 1, \\ 1, & x > 1. \end{cases}$$

(a) What is $P(X \ge 1/2)$?

(b) What is $P(-1/2 \le X < 3/4)$?

(c) What is $P(|X| > 1/2)$?

(d) What is the value of a such that $P(X < a) = 0.8$?

(e) Find the PDF $f_X(x)$ of X.

Solution. (a) $1/4$ (b) $5/8$ (c) $1/2$ (d) $a = 0.6$

Continuous Random Variables

Exercise 15

The cumulative distribution function of the continuous random variable V is

$$F_V(v) = \begin{cases} 0, & v \leq -5, \\ c(v+5)^2, & -5 < v \leq 7, \\ 1, & v > 7. \end{cases}$$

(a) What is c?

(b) What is $P(V \geq 4)$?

(c) $P(-3 \leq V < 0)$?

(d) What is the value of a such that $P(V \geq a) = 2/3$?

Solution. (a) $1/144$ (b) $63/144$ (c) $21/144$ (d) 1.928

Continuous Random Variables

Exercise 16

The random variable X has probability density function

$$f_X(x) = \begin{cases} cx, & 0 \leq x \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

Use the PDF to find

(a) The constant c,

(b) $P(0 \leq X \leq 1)$,

(c) $P(-1/2 \leq X \leq 1/2)$,

(d) The CDF $F_X(x)$.

Solution. (a) $c = 1/2$ (b) $1/4$ (c) $1/16$ (d) $F_X(x) = \begin{cases} 0, & x \leq 0, \\ x^2/2, & 0 < x \leq 2, \\ 1, & x > 2. \end{cases}$

Continuous Random Variables

Exercise 17

The cumulative distribution function of the continuous random variable X is

$$F(x) = \begin{cases} 0, & x \le -a \\ A + B \arcsin \dfrac{x}{a}, & -a < x \le a \\ 1, & x > a. \end{cases}$$

(a) What are A and B?

(b) What is the PDF $f_X(x)$?

Continuous Random Variables

Exercise 18

The cumulative distribution function of the continuous random variable X is $F(x) = a + b \arctan x$, $-\infty < x < +\infty$.

(a) What are a and b?

(b) What is the PDF $f_X(x)$ of X?

(c) What is $P(-1 < X < 1)$?

Solution. (a) $1/2 \quad 1/\pi$.

Continuous Random Variables

Exercise 19

The cumulative distribution function of the continuous random variable X is

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \arctan \frac{x}{2}.$$

What is the value of x_1 such that $P(X > x_1) = 1/4$?

Exercise 20

The continuous random variable X has probability density function

$$f(x) = \begin{cases} k \sin 3x, & x \in \left(0, \frac{\pi}{3}\right), \\ 0, & x \notin \left(0, \frac{\pi}{3}\right). \end{cases}$$

Use the PDF to find

(a) the constant k, (b) $P(\pi/6 \leq X < \pi/3)$, (c) the CDF $F_X(x)$.

Continuous Random Variables

Exercise 21

The random variable X has the probability density function

$$f_X(x) = \begin{cases} k(30-x), & x \in (0,30), \\ 0, & x \notin (0,30). \end{cases}$$

Let $Y := \max\{20, X\}$. Find the expected value of the random variable Y.

Important Probability Distributions

Exercise 22

A rat maze consists of a straight corridor, at the end of which is a branch; at the branching point the rat must either turn right or left. Assume 10 rats are placed in the maze, one at a time.

(a) If each is choosing one of the two branches at random, what is the distribution of the number that turn right?

(b) What is the probability at least 9 will turn the same way?

Exercise 23

The number of phone calls at a post office in any time interval is a Poisson random variable. A particular post office has on average 2 calls per minute.

(a) What is the probability that there are 5 calls in an interval of 2 minutes?

(b) What is the probability that there are no calls in an interval of 30 seconds?

(c) What is the probability that there are no less than one call in an interval of 10 seconds?

Important Probability Distributions

Exercise 24

A light bulb has a lifetime which is exponential with parameter $\lambda = 0.5/\text{year}$.

(a) If the bulb is replaced with a new one when it fails what is the expected number of bulbs used during 8 years?

(b) If the bulb is still working after 2 years then it is replaced by a new one. Let Y be the time that the bulb is in use. Find the cumulative distribution function $F_Y(y)$ of Y.

(c) Find $E(Y)$.

Important Probability Distributions

Exercise 25

Suppose the number of customers, arriving at a market is modeled as a Poisson Process with mean arrival rate $\lambda = 4/\text{hour}$.

(a) If exactly 2 customers arrive during 9:00-9:30 what is the probability that at least 8 customers arrive during 9:00-10:30?

(b) If less than 20 customers arrive during 9:00-17:30, the market is assumed to be nonprofitable. What is the probability that exactly 1 day will be profitable in a week? (Assume that market is open 6 days in a week).

Important Probability Distributions

Exercise 26

Starting at 5:00 am, every half hour there is a flight from San Francisco airport to Los Angeles International Airport. Suppose that none of these planes sold out and that they alwayshave room for passengers. A person who wants to fly LA arrives at the airport at a random time between 8:45–9:45 am. Find the probability that she waits at most 10 minutes and at least 15 minutes.

Important Probability Distributions

Exercise 27

Let X be an exponential random variable with parameter and define $Y = [X]$, the largest integer in X, (ie. $[x] = 0$ for $0 \leq x < 1$, $[x] = 1$ for $1 \leq x < 2$ etc.)

(a) Find the probability function for Y.

(b) Find $E(Y)$.

(c) Find the distribution function of Y.

(d) Let Y represent the number of periods that a machine is in use before failure. What is the probability that the machine is still working at the end of 10th period given that it does not fail before 6th period?

Important Probability Distributions

Exercise 28

X is an exponential random variable with the PDF $f_X(x)$ is

$$f_X(x) = \begin{cases} 5e^{-5x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

(a) What is $E(X)$?

(b) What is $P(0.4 < X < 1)$?

Important Probability Distributions

Exercise 29

X is a Gaussian random variable with $E(X) = 0$ and $\sigma_X = 0.4$.

(a) What is $P(X > 3)$?

(b) What is the value of c such that $P(3 - c < X < 3 + c) = 0.9$?

Exercise 30

X is a Gaussian random variable with $E(X) = 0$ and $P(|X| \leq 10]) = 0.1$. What is the standard deviation σ_X?